Random Forests

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CM22009 – Machine Learning

This lecture adapts material originally prepared by Dr. Andy Barnes (with permission)

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Recap: Decision Trees

- Split criteria: choose a split with largest **information gain**

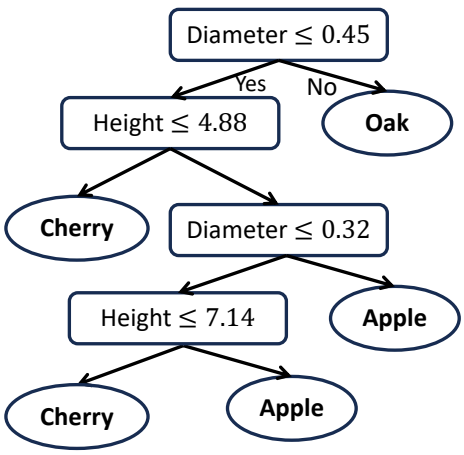
For categorical data: $\max_{a} IG(S, a)$

$$IG(S, a) = E(S) - \sum_{v \in \mathcal{A}} \frac{|S_v|}{|S|} E(S_v)$$

For continuous data: $\max_{j,\theta} IG(x_j, \theta)$

$$IG(x_j, \theta) = E(S) - \left(\frac{|L|}{|S|} E(L) + \frac{|R|}{|S|} E(R)\right)$$

$$E(S) = -\sum_{c=1}^{C} p_c \log_2(p_c)$$



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Where are we?

Week No.	Lecture A	Lecture B
1 (29/09/2025)	Intro. Supervised Learning	Logistic Regression
2 (06/10/2025)	Nearest-Neighbours	Support Vector Machines
3 (13/10/2025)	Decision Trees	Random Forests
4 (20/10/2025)	Naïve-Bayes	Data Engineering
5 (27/10/2025)		
6 (03/11/2025)	Reading Week	

Gradient Boosting

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Learning Objectives

Describe the principles behind random forests.

Build and evaluate random forests.

Discuss the advantages and disadvantages of random forests.

Explain different types of ensemble learning.

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Random Forests

An ensemble of decision trees

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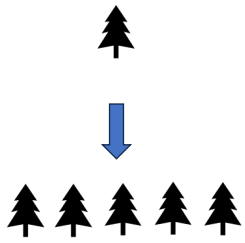
Why go beyond a single decision tree?

✓Easy to interpret

✓Flexible

Prone to overfitting

Unstable (small changes -> very different tree)



Each tree is a "weak" learner, capturing different patterns

Goal: Reduce variance and improve stability by combining many trees

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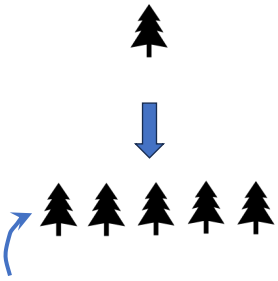
Random Forests:

Build several decision trees, then averages or votes their predictions.

Two main sources of randomness (for diversity):

Bootstrap Sampling (Bagging) — Each trained on a random subset of data (with replacement)

Random Feature Selection — Each split considers only a random subset of features



Each tree is a "weak" learner, capturing different patterns

Q: How could we build many decision trees on the same dataset?

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Constructing Random Forests

1 Create a bootstrapped dataset.

- **A** Select a random row from training set.
- **B** Append it to a new dataset.

Repeat until the new dataset is the same size as the training dataset.

Flood Forecasting(Training)			
Date	**Type**	**Rainfall**	**Flood**
21/02/2023	A	207	True
02/02/2023	B	103	False
02/02/2023	C	95	False
19/01/2023	A	195	True

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Constructing Random Forests

1 Create a bootstrapped dataset.

A Select a random row from training set.

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Repeat until the new dataset is the same size as the training dataset.

Note: Some samples might appear multiple times. Not every sample is selected.

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Constructing Random Forests

2 Build a decision tree on this dataset.

* At each node only consider a random selection of m features. ($m < D$)

N_1

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Q: How do we choose m?

Typically $m = \sqrt{D}$ features for classification
Typically $m = D/3$ features for regression

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Constructing Random Forests

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N_1

N_2

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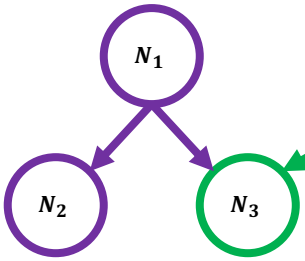

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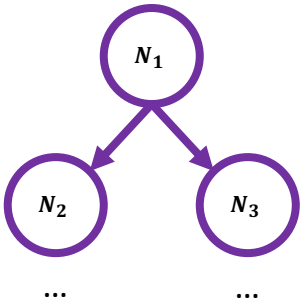

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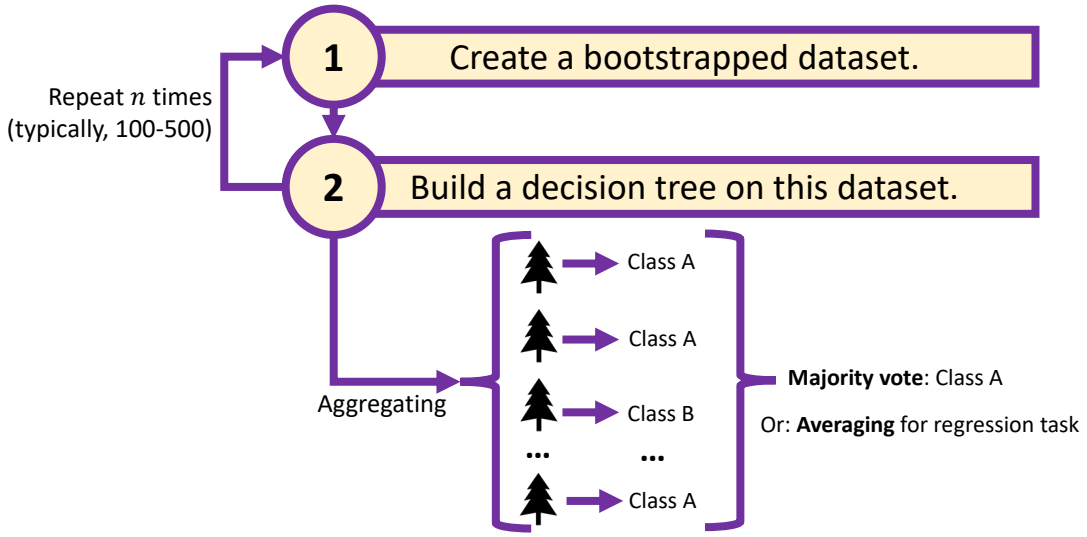
Constructing Random Forests



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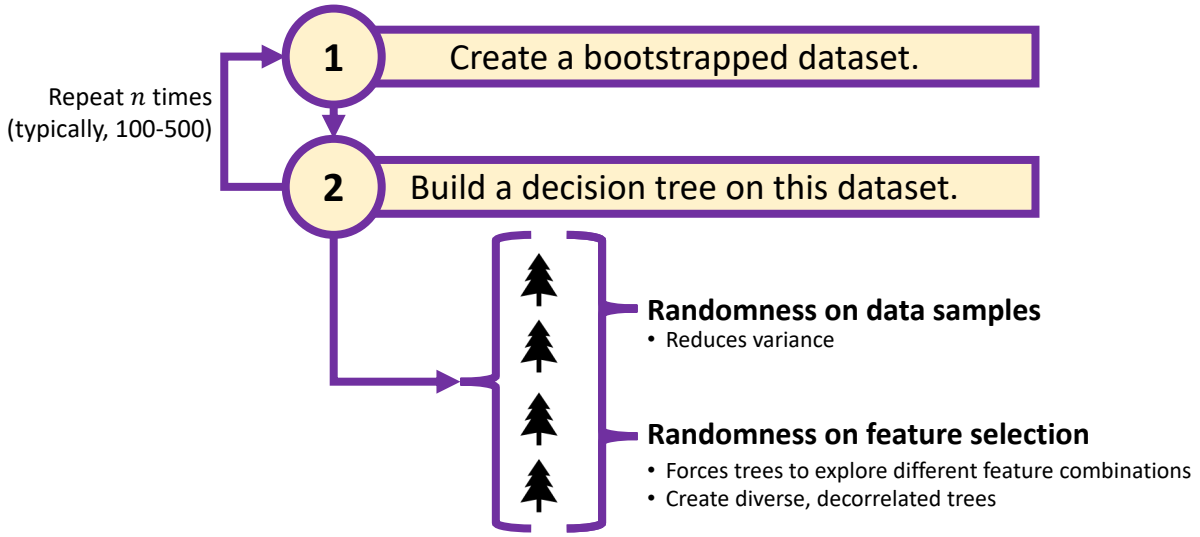
Constructing Random Forests



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Out-of-Bag (OOB) Error

- Bootstrapped dataset: a random subset (with replacement) of all training samples
- Each tree only used a subset of samples (~63%)
- Around 37% not in bootstrap (called **OOB samples**)→ a built-in validation set
- Can estimate error on these OOB samples (called **OOB error**) without separate validation set

In Random forests, we often use the OOB error instead of a separate validation error to tune hyperparameters.

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Advantages	Disadvantages
Reduces overfitting	**Lack of interpretability**
Minimal data preparation (No scaling needed, handle missing values)	**Computationally expensive and high memory usage**
Robustness to noise/outliers	**Longer prediction time**
Feature importance estimation	

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Ensemble Learning

wisdom of the crowd

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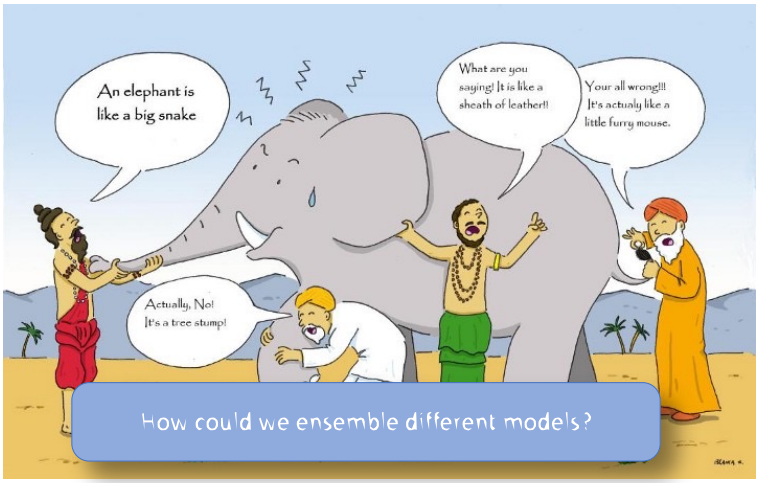
Ensemble Learning



- Intuition: wisdom of the crowd

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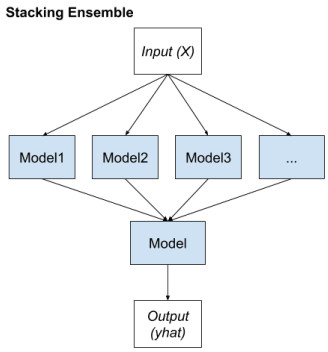
Ensemble learning

- Opt 1: **Stacking**
  - Different machine learning algorithms for each ensemble member.
  - A **meta-learner** to learn how to best combine predictions.



Img from: https://machinelearningmastery.com/tour-of-ensemble-learning-algorithms

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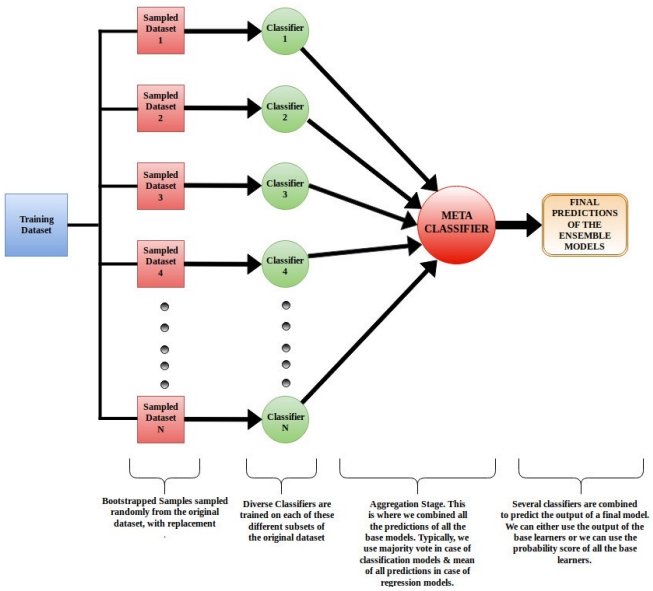
Ensemble learning

- Opt2: **Bagging** (short for Bootstrap Aggregating)
- **Bagging** is a method of training multiple models using a random subset of the training data with replacement.



Img from: https://thmachne.blogspot.com/2021/02/bagging-machine-learning-ppt.html

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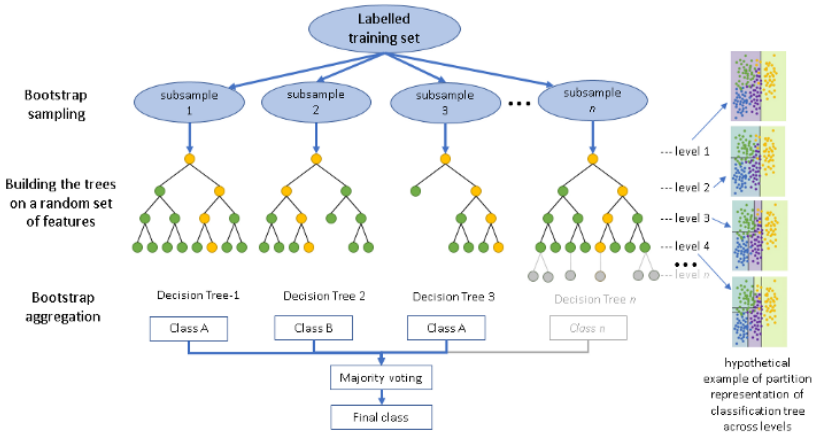
Bagging

- **Random Forest** = Bagging + Decision Tree + Random set of features



Img from: https://medium.com/nerd-for-tech/random-forest-sturdy-algorithm-d60b9f9140d4

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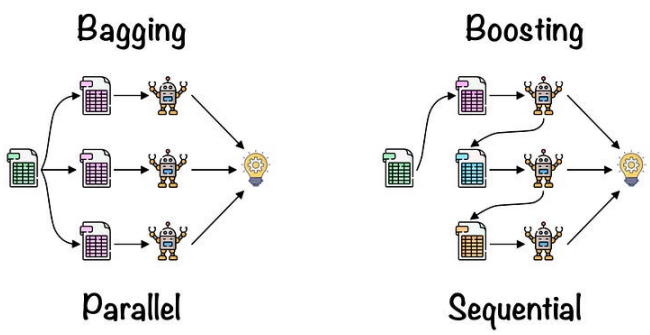
Ensemble learning

- Opt 3: **Boosting**
  - "Learning from mistakes"



Principle: Build a strong model from sequential models, where each model improves or corrects the mistake made in the previous ones.

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Boosting

- AdaBoost (Adaptive Boosting):
  - Decision Trees (stumps) + Boosting (adjust sample weights)
- Gradient Boosting
  - Decision Trees (shallow) + Boosting (fit to residual errors)

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Feedback!

If you have any feedback, use the QR code or link to anonymously report it.

I'm open to any comments/feedback.

SCAN ME

https://forms.gle/AyTC145yPeFVMiQw8

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